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Dimostrare che il seguente linguaggio NON è CFL:

$$A = \{ w \# v \mid w, v \in \{0,1\}^* \text{ dove } w \text{ è la codifica binaria di un numero } m \geq 1 \text{ e } v \text{ è la codifica binaria del numero } m+1 \}$$

SVLG

$$\begin{bmatrix} 110 \# 111 \\ m \quad m+1 \end{bmatrix} \in L$$

$$\begin{bmatrix} 1010 \# 1011 \end{bmatrix} \in L$$

$$\begin{array}{c|c} 7 & 8 \\ 111 & 1000 \end{array}$$

$$\begin{array}{c|c} p & p+1 \\ 101 & 10 \end{array}$$

00	0
01	1
10	2
11	3
100	4
101	5
110	6
111	7
1000	8
1001	9
1010	10
1011	11
1100	12
1101	13
1110	14
1111	15

$$S = \overset{p}{10} \mid \overset{p}{11} = 110 \mid 111 = 1110 \mid 1111$$

$$1) S = \underbrace{1111 \dots 1}_p 0 \mid \underbrace{1111 \dots 1}_p 1$$

$v \times y$

$$\begin{bmatrix} v=1 & v=x^+ \\ x=1 & x=1^+ \\ y=1 & y=1^+ \end{bmatrix} \text{ UP: } 1^S 0 \mid 1^p 1, S \neq p, \notin L$$

$$2) S = \underbrace{1111 \dots 1}_p 0 \mid \underbrace{1111 \dots 1}_p 1$$

$v \times y$

$$\begin{bmatrix} v=1 & v=x^+ \\ x=1 & x=1^+ \\ y=1 & y=1^+ \end{bmatrix} \text{ UP: } 1^p 0 \mid 1^S 1, S > p, \notin L$$

$$3) S = \underbrace{1111 \dots 1}_p 0 \mid \underbrace{1111 \dots 1}_p 1$$

$$\begin{bmatrix} v=1 & v=x^+ \\ x=1 & x=1^+ \\ y=0 & y=0 \end{bmatrix} \text{ UP: } \begin{array}{c|c} S & 1 \\ 1 & 1 \end{array} 0 \mid 1^p 1 \rightarrow \begin{array}{c|c} q & q \\ 1 & 0 \end{array} \mid 1^p 1, \text{ aumento il numero di zeri! } \notin L$$

$$3) S = \underbrace{1111 \dots 1}_p 0 \mid \underbrace{1111 \dots 1}_p 1$$

vxy

$$\begin{bmatrix} v=1 & v=1^+ \\ x=1 & x=1^+ \\ y=1 & y=1 \end{bmatrix}$$

$$UP: 1^p 0 \mid 1^p 1 \rightarrow 1^p 0 \mid 1^q 1$$

→ aumento il numero di 0's.
 $\notin L$.

$$4) S = \underbrace{1111 \dots 1}_p 0 \mid \underbrace{1111 \dots 1}_p 1$$

vxy

$$1^p 0 \mid 1^p 1$$

$$\begin{bmatrix} v=1 & v=1^+ \\ x=0 & x=0 \\ y=1 & y=1^+ \end{bmatrix}$$

$$UP: 1^S 0 \mid 1^S 1 \in L$$

questa stringa non va bene X.

$$S = 1^p 0 1^p \# 1^{p+1} 0$$

$$1) S = \underbrace{1111 \dots 1}_p 0 \underbrace{1111 \dots 1}_p \# \underbrace{11111 \dots 1}_{p+1} \underbrace{0000 \dots 0}_p$$

vxy

$$\begin{bmatrix} v=1 & v=1^+ \\ x=0 & x=0 \\ y=1 & y=1^+ \end{bmatrix}$$

$$UP: 1^S 0 1^S \# 1^{p+1} 0 \rightarrow$$

$$S > p > p+1, \notin L$$

↓
dimensione arbitraria!

$$2) S = \underbrace{1111 \dots 1}_p 0 \underbrace{1111 \dots 1}_p \# \underbrace{11111 \dots 1}_{p+1} \underbrace{0000 \dots 0}_p$$

vxy

$$\begin{bmatrix} v=1 & v=1^+ \\ x=10 & x=1^+ 0^+ \\ y=0 & y=0^+ \end{bmatrix}$$

$$UP: S = 1^p 0 1^p \# 1^S 1^q 0^{q-1} S$$

$1^p 0 1^p \# 1^q 0^{q-1}$

$$\underline{q \neq p+1}$$

$q > q-1$

$$3) S = \underbrace{1111 \dots 1}_p \underbrace{01111 \dots 1}_p \# \underbrace{11111 \dots 1}_{p+1} \underbrace{0000 \dots 0}_p$$

xy

$$\left[\begin{array}{l|l} v=1 & v=1^+ \\ x=\# & x=\# \\ y=1 & y=1^+ \end{array} \right] \text{ up: } 1^p 0 1^S \# 1^{S+1} 0^p$$

$$\begin{array}{l} S+1 \geq P+1 \\ \hline S+1 \neq P+1 \\ S \in \text{ARBITRARY} \notin L \end{array}$$

L IS NOT CFL!